GEOMETRY AND MEASURES

Questions 1 – 15 · Worked Solutions in Question and Answer Form

Compiled Scenario Questions of Mathematics Mock Papers 2026

UCE Mathematics (Senior Four) · Integration / scenario items · Revision guide for UNEB candidates

No.	Topic of the scenario	Mock board
1	Circle properties and bearing	WAKISSHA
2	Foreign exchange, interest and hire purchase	WAKISSHA
3	Circle properties (intersecting circles)	KAMSSA
4	Areas and volumes of solids (cones)	KAMSSA
5	Bearing, trigonometry and kinematics	WAKATA
6	Nets, volumes of solids and compound interest	WAKATA
7	Matrices of transformation and construction	COUHEIA
8	Areas and volumes of solids; angles of elevation	COUHEAI
9	Similar solids, areas, currency exchange and depreciation	SHURE
10	Vectors, reflection and areas of 2D shapes	SHURE
11	Lines and planes in 3D; rotation	UTES
12	Circle properties and mortgage	UTES
13	Vector geometry, bearing, surface area and volume	UMTA
14	Bearing, construction and taxation (PAYE)	UMTA
15	Surface area and transformations	ACEITEKA

Assumptions used in this guide (check them against your school copy of the paper)

- Calculator value of π is used unless the question fixes it (Q3 uses 3.14; Q9 uses 22/7).
- Q2: the interest rates are per annum. Q5: the bearing "N50°S" is read as **N50°E (050°)**.
- Q6: the 8 equal instalments are shared equally from the compound amount. Q9: volumes are in m^3 (the paper prints cm^3).
- Q13: 35° is the angle between the slant side of the cone and its horizontal base.

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On a farm, a man ties a bull to a point **A** with a rope of length 10 m and a cow to a point **B** with a rope of length 8 m, where AB = 12 m. Each animal grazes in a circular region centred at its tying point. A worker moves from the bull a distance of 15 m on a bearing of 060° to collect a jerrycan, then 25 m on a bearing of 150° to the well. On coming back he uses the direct route to the bull.

Task: (a) Determine the area of the region common to both grazing animals. (b) Find the direct distance the worker will walk from the well to the bull.

(a) Area of the region common to both grazing regions

The circles meet at C and D. In triangle ABC the sides are AC = 10 m, BC = 8 m and AB = 12 m. The common region is made of **two segments** cut off by the common chord CD (the centres A and B lie on opposite sides of CD).

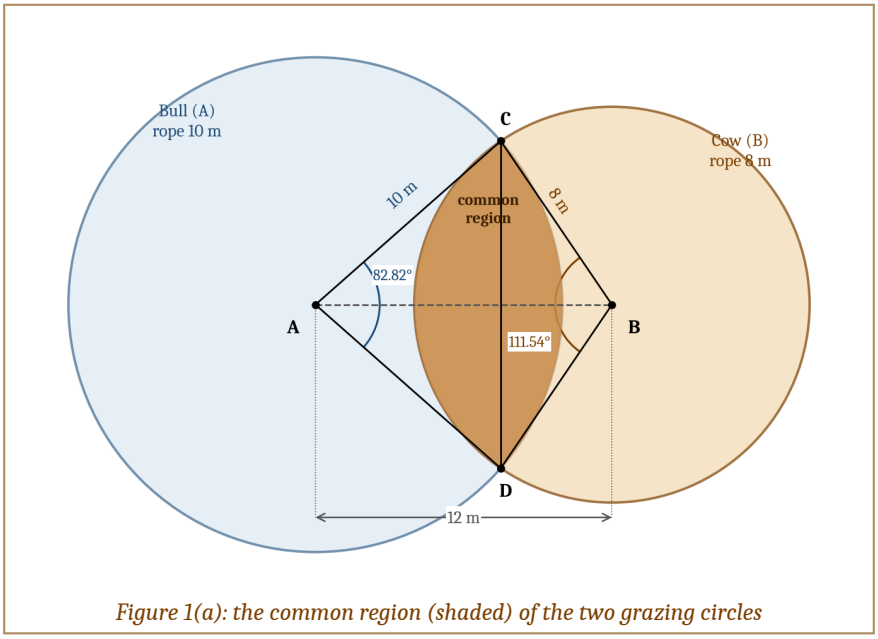


Figure 1(a): the common region (shaded) of the two grazing circles

Step 1 – angles at the centres (cosine rule).

$$\cos \angle CAB = \frac{10^2 + 12^2 - 8^2}{2 \times 10 \times 12} = \frac{180}{240} = 0.75 \Longrightarrow \angle CAB = 41.41°$$

So $\angle CAD = 2 \times 41.41° =$ **82.82°**.

$$\cos \angle CBA = \frac{8^2 + 12^2 - 10^2}{2 \times 8 \times 12} = \frac{108}{192} = 0.5625 \Longrightarrow \angle CBA = 55.77°$$

So $\angle CBD = 2 \times 55.77° =$ **111.54°**.

Step 2 – segment of the bull's circle (radius 10 m, angle 82.82°): segment = sector − triangle

$$= \frac{82.82}{360} \times \pi \times 10^2 - \tfrac{1}{2} \times 10 \times 10 \times \sin 82.82° = 72.27 - 49.61 = 22.67 \text{ m}^2$$

Step 3 – segment of the cow's circle (radius 8 m, angle 111.54°):

$$= \frac{111.54}{360} \times \pi \times 8^2 - \tfrac{1}{2} \times 8 \times 8 \times \sin 111.54° = 62.30 - 29.76 = 32.53 \text{ m}^2$$

Step 4 – add the two segments: $22.67 + 32.53 = 55.20$ m^2.

∴ The area common to both grazing regions is ≈ 55.2 m^2.

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(b) Direct distance from the well to the bull

Let the bull be at A, the jerrycan at P and the well at W. The path A → P is on 060° and P → W is on 150°. At P the back-bearing of PA is 060° + 180° = 240°. Hence ∠APW = 240° − 150° = **90°**, so triangle APW is right-angled at P.

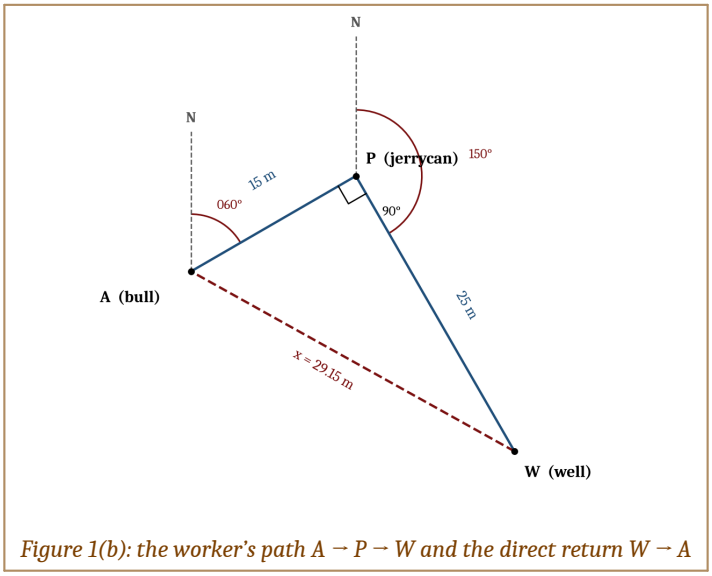


Figure 1(b): the worker's path A → P → W and the direct return W → A

$$AW^2 = AP^2 + PW^2 = 15^2 + 25^2 = 225 + 625 = 850 \implies AW = \sqrt{850} = 29.15 \text{ m}$$

Direction of the return: $\angle PAW = \tan^{-1}(25 \div 15) = 59.04°$, so AW is on 060° + 59.04° = 119.04°, and the return W → A is on 119.04° + 180° ≈ **299°**.

∴ The worker walks 29.15 m (≈ 29.2 m) from the well to the bull, on a bearing of about 299°.

> **Key idea:** Common area of two intersecting circles = sector − triangle for each circle, then add. Whenever two bearings meet at a turning point, find the angle by using the back-bearing.

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QUESTION 2	FOREIGN EXCHANGE, INTEREST AND HIRE PURCHASE	WAKISSHA

Bob wants to start a business worth Shs 7,000,000. His friend in the USA sent him US$ 1,400 (exchange rate $1 = UGX 3,800). He borrows the remaining amount from the cheaper of two banks. **Bank A**: simple interest of 20% for 24 months. **Bank B**: compound interest of 14% for 24 months.

Alex wants a car whose cash price is UGX 32,000,000. **(i)** pay 25% of the cash value, then monthly instalments of Shs 1,125,000 for 2 years; **(ii)** pay ⅛ of the cash value, then Shs 315,000 weekly for 104 weeks.

Task: (a) Advise Bob on the cheapest bank. (b) Help Alex to determine the less costly hire-purchase alternative.

(a) The cheaper bank for Bob

Money already available: US$ 1,400 × 3,800 = Shs 5,320,000.

Amount to borrow: 7,000,000 − 5,320,000 = **Shs 1,680,000**.

Bank A (simple interest, 20% per annum, 2 years):

$$I = \frac{P \times R \times T}{100} = \frac{1{,}680{,}000 \times 20 \times 2}{100} = \text{Shs } 672{,}000 \implies \text{amount to repay} = \text{Shs } 2{,}352{,}000$$

Bank B (compound interest, 14% per annum, 2 years):

$$A = P(1 + R/100)^n = 1{,}680{,}000 \times (1.14)^2 = 1{,}680{,}000 \times 1.2996 = \text{Shs } 2{,}183{,}328$$

Interest = 2,183,328 − 1,680,000 = Shs 503,328.

Bank	Interest to pay	Total to repay
A (simple 20%)	Shs 672,000	Shs 2,352,000
B (compound 14%)	Shs 503,328	Shs 2,183,328

∴ Bank B is cheaper: Bob saves 672,000 − 503,328 = Shs 168,672. He should borrow from Bank B.

(b) The less costly hire-purchase option for Alex

	Option (i)	Option (ii)
Deposit	25% × 32,000,000 = 8,000,000	⅛ × 32,000,000 = 4,000,000
Instalments	1,125,000 × 24 months = 27,000,000	315,000 × 104 weeks = 32,760,000
Hire-purchase price	8,000,000 + 27,000,000 = **35,000,000**	4,000,000 + 32,760,000 = **36,760,000**
Extra over cash price	35,000,000 − 32,000,000 = 3,000,000	36,760,000 − 32,000,000 = 4,760,000

∴ Option (i) is less costly; it is cheaper than option (ii) by 36,760,000 − 35,000,000 = Shs 1,760,000.

Key idea: Compare the TOTAL cost: deposit + (instalment × number of instalments). For compound interest use $A = P(1 + R/100)^n$; simple interest uses $I = PRT/100$.

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QUESTION 3 CIRCLE PROPERTIES – INTERSECTING CIRCLES KAMSSA

Your uncle wants a design of intersecting circles in his compound: one circle of radius 2.0 m and another of radius 1.5 m, with a distance of 1.8 m between the two points of intersection. Pavers are to be placed in the area shared by the two circles. Each paver measures 4.0 cm by 8.2 cm and costs Shs 250.

Task: Determine the total cost of pavers needed. [Take π = 3.14]

The common chord CD = 1.8 m, so half of it is 0.9 m. The shared area (the pathway) is the sum of the two segments on either side of CD.

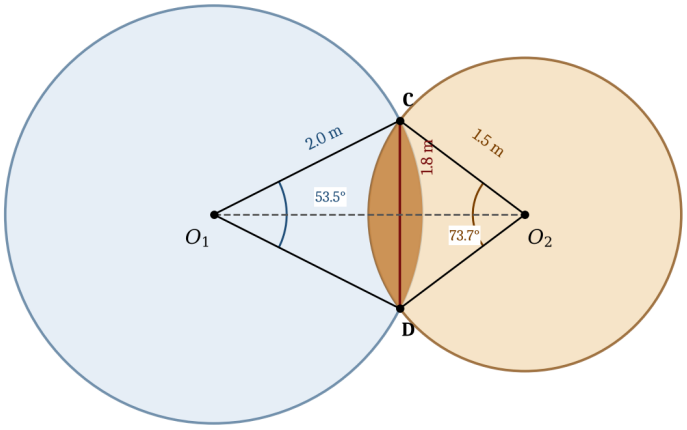


Figure 3: the shared region (shaded) between circles O_1 (radius 2.0 m) and O_2 (radius 1.5 m)

Step 1 – angles at the centres.

$$\sin ½\angle CO_1D = \frac{0.9}{2.0} = 0.45 \implies ½\angle CO_1D = 26.74° \implies \angle CO_1D = 53.49°$$

$$\sin ½\angle CO_2D = \frac{0.9}{1.5} = 0.6 \implies ½\angle CO_2D = 36.87° \implies \angle CO_2D = 73.74°$$

Step 2 – segment of the larger circle (r = 2.0 m):

$$= \frac{53.49}{360} \times 3.14 \times 2^2 - ½ \times 2^2 \times \sin 53.49° = 1.8662 - 1.6075 = 0.2587 \text{ m}^2$$

Step 3 – segment of the smaller circle (r = 1.5 m):

$$= \frac{73.74}{360} \times 3.14 \times 1.5^2 - ½ \times 1.5^2 \times \sin 73.74° = 1.4471 - 1.0800 = 0.3671 \text{ m}^2$$

Step 4 – area of the shared region: $0.2587 + 0.3671 = 0.6258 \text{ m}^2 = 0.6258 \times 10{,}000 =$ **6,258 cm²**.

Step 5 – the pavers. Area of one paver = $4.0 \times 8.2 = 32.8 \text{ cm}^2$.

$$\text{Number of pavers} = \frac{6{,}258}{32.8} = 190.8 \implies 191 \text{ pavers (round up so that the area is fully covered)}$$

Cost = 191 × Shs 250 = Shs 47,750.

∴ **The total cost of the pavers is Shs 47,750.**

Key idea: Convert units first ($1 \text{ m}^2 = 10{,}000 \text{ cm}^2$). Always round the number of tiles/pavers UP because you cannot buy part of a paver.

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Your uncle makes ice-cream cones of carrying capacity 32 cm^3 and vertical height 4 cm. A cone costs him Shs 50 to make and he sells it at Shs 200. The selling price is directly proportional to the capacity, and the cost varies directly as the total material needed. A new client orders 250 similar cones, each of capacity 256 cm^3.

Task: Determine the (a) diameter and vertical height of the new cone; (b) amount of material needed for the order and the cost he is likely to incur; (c) price to charge for each cone and the profit from the order. [Use π = 3.14]

(a) Diameter and height of the new cone

Old cone: $V = \frac{1}{3}\pi r^2 h$, so

$$32 = \frac{1}{3} \times 3.14 \times r^2 \times 4 \implies r^2 = 96 \div 12.56 = 7.643 \implies r = 2.765 \text{ cm}$$

Scale factor. The cones are similar, so volume ratio = 256 ÷ 32 = 8 = k^3, giving a linear scale factor $k = \sqrt[3]{8} = \mathbf{2}$.

New height = 2 × 4 = 8 cm, new radius = 2 × 2.765 = 5.53 cm, new diameter = 11.06 cm

Check: $V = \frac{1}{3} \times 3.14 \times 5.53^2 \times 8 = 256.1\ cm^3$ ✓.

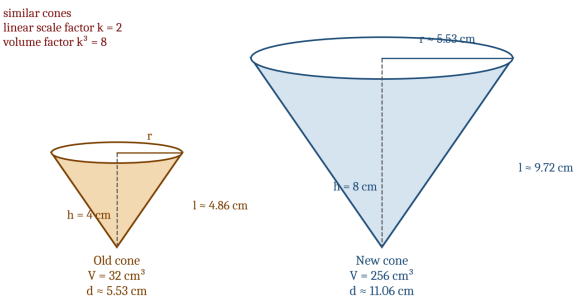


Figure 4: the old cone and the new (similar) cone

∴ The new cone has a vertical height of 8 cm and a diameter of ≈ 11.06 cm.

(b) Material needed and the cost

An ice-cream cone is open at the top, so the material is the **curved surface area** πrl.

$$\text{Old cone: } l = \sqrt{(r^2 + h^2)} = \sqrt{(7.643 + 16)} = 4.862 \text{ cm} \implies \text{CSA} = 3.14 \times 2.765 \times 4.862 = 42.21\ cm^2$$

Areas of similar solids are in the ratio $k^2 = 4$, so new CSA = 4 × 42.21 = **168.84 cm^2** per cone.

Material for 250 cones = 250 × 168.84 = 42,210 cm^2 (≈ 4.22 m^2).

Cost ∝ material: cost of a new cone = 50 × 4 = Shs 200, so the cost of the order = 250 × 200 = Shs 50,000.

∴ He needs 42,210 cm^2 of material and will incur a cost of Shs 50,000.

(c) Selling price and profit

Price ∝ capacity: the capacity is 8 times larger, so the price is 8 × 200 = **Shs 1,600** per cone.

$$\text{Revenue} = 250 \times 1{,}600 = \text{Shs } 400{,}000 \implies \text{Profit} = 400{,}000 - 50{,}000 = \text{Shs } 350{,}000$$

∴ Each cone is sold at Shs 1,600 and the profit on the order is Shs 350,000.

Key idea: For similar solids: lengths × k, areas × k^2, volumes × k^3. Cost follows area; price follows volume (as stated in the question).

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A delivery truck leaves a warehouse W at 7:00 a.m. and travels 150 km on a bearing of N50°E to the first retail outlet O_1, arriving at 8:45 a.m. The staff spend 45 minutes unloading. The truck then leaves for the second outlet O_2, which lies on a bearing of 150° from the warehouse, travelling at 80 km/h for 3 hours. The staff spend 1 hour unloading. The truck returns directly from O_2 to W at 60 km/h and must be back by 5:30 p.m.

Task: (a) Find the distance of the direct route from the second outlet back to the warehouse. (b) Advise whether the truck will be back by 5:30 p.m.

Interpretation used: the first bearing is read as N50°E (= 050°). Then the angle between WO_1 and WO_2 is 150° − 50° = 100°.

(a) Distance from O_2 to W

Distance O_1 to O_2 = speed × time = 80 × 3 = **240 km**. Let WO_2 = d. In triangle WO_1O_2: WO_1 = 150 km, O_1O_2 = 240 km, $\angle O_1WO_2$ = 100°.

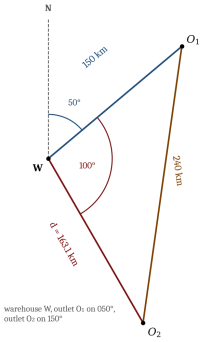


Figure 5: the warehouse W and the two outlets O_1 and O_2

Cosine rule (to find the unknown side d):

$$240^2 = 150^2 + d^2 - 2(150)(d)\cos 100°$$

$$57{,}600 = 22{,}500 + d^2 + 52.09d \implies d^2 + 52.09d - 35{,}100 = 0$$

$$d = \frac{-52.09 + \sqrt{(52.09^2 + 4 \times 35{,}100)}}{2} = \frac{-52.09 + 378.3}{2} = 163.1 \text{ km}$$

∴ The direct route from the second outlet to the warehouse is 163.1 km.

(b) Will the truck reach the warehouse by 5:30 p.m.?

Stage	Working	Time
Leaves warehouse	—	7:00 a.m.
Arrives at O_1	150 km in 1 h 45 min (given)	8:45 a.m.
Unloading at O_1	45 minutes	leaves 9:30 a.m.
Travel $O_1 \to O_2$	3 hours at 80 km/h (240 km)	12:30 p.m.
Unloading at O_2	1 hour	leaves 1:30 p.m.
Return $O_2 \to$ W	time = 163.1 ÷ 60 = 2.72 h = 2 h 43 min	arrives ≈ 4:13 p.m.

∴ The truck reaches the warehouse at about 4:13 p.m., which is 1 hour 17 minutes before 5:30 p.m. The manager should be advised that the truck will be back in time and no overtime costs arise.

Key idea: Draw a sketch with the north line at the starting point. Angle between two bearings from the same point = difference of the bearings. Time = distance ÷ speed.

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QUESTION 6	NETS, VOLUMES OF SOLIDS AND COMPOUND INTEREST	WAKATA

A youth group packages gifts in pyramid-shaped boxes. The net has a square base of side 12 cm and four congruent isosceles triangular faces; the distance from the apex of a triangular face to the midpoint of its base is 40 cm. The space is sufficient if it exceeds 1500 cm^3. The group also takes a loan of UGX 19.45 million at a compound interest of 10.5% per annum, to be repaid in two years in 8 equal instalments.

Task: (a) Determine the vertical height of the pyramid. (b) Advise whether it provides enough space. (c) Determine the amount paid in each of the 8 equal instalments.

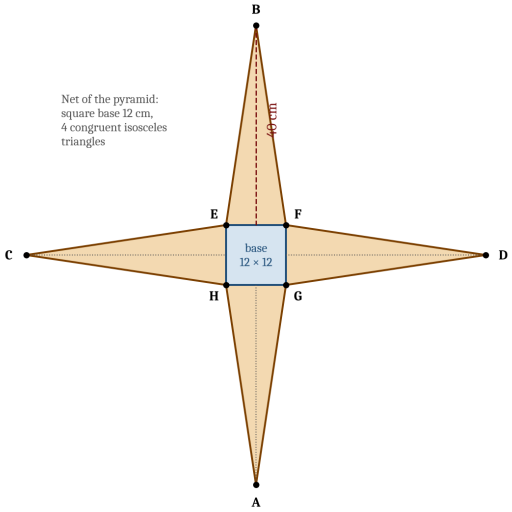


Figure 6(a): the net

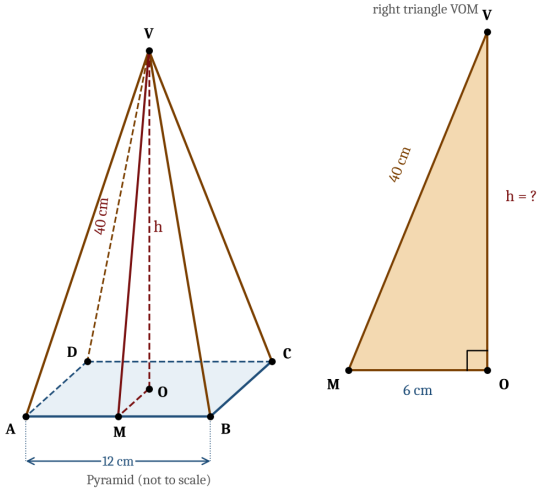


Figure 6(b): the pyramid and triangle VOM

(a) Vertical height of the pyramid

Let O be the centre of the base and M the midpoint of a base edge. Then OM = 12 ÷ 2 = 6 cm and the slant height VM = 40 cm. Triangle VOM is right-angled at O.

$$h^2 = VM^2 - OM^2 = 40^2 - 6^2 = 1600 - 36 = 1564 \implies h = \sqrt{1564} = 39.55 \text{ cm}$$

∴ The vertical height is 39.55 cm.

(b) Is there enough space?

$$V = \tfrac{1}{3} \times \text{base area} \times \text{height} = \tfrac{1}{3} \times (12 \times 12) \times 39.55 = \tfrac{1}{3} \times 144 \times 39.55 = 1{,}898.3 \text{ cm}^3$$

∴ Since 1,898.3 cm^3 > 1,500 cm^3, the pyramid package provides enough space for the gifts.

(c) Loan instalments

Amount after 2 years at compound interest:

$$A = P(1 + R/100)^n = 19{,}450{,}000 \times (1.105)^2 = 19{,}450{,}000 \times 1.221025 = \text{Shs } 23{,}748{,}936.25$$

Equal instalments:

$$\text{Instalment} = \frac{23{,}748{,}936.25}{8} = \text{Shs } 2{,}968{,}617$$

∴ Each of the 8 instalments is ≈ Shs 2,968,617 (total interest paid ≈ Shs 4,298,936).

Key idea: In a net, slant height (apex to the midpoint of the base edge) is NOT the edge length. Use half the base to form the right-angled triangle that gives the vertical height.

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A KCCA graphics team works on a triangular logo A(3, 1), B(5, 1), C(5, 3) of area 2 cm^2. The triangle is reflected in the line y = x and then enlarged by scale factor 2. The team wants the matrix that returns the final image to its original position, the coordinates at each stage and the areas. Another team constructs triangle ABC with BC = 8 cm, AC = 12 cm and ∠ABC = 105°, then locates D equidistant from A and B, on the same side of AB as C, with ∠ADC = 90°.

Task: (a) Find the matrices and the coordinates of the images. (b) Determine the area of the transformed triangles. (c) Show the exact position of D and find the lengths (i) AB (ii) AD (iii) DC.

(a) Matrices of transformation and the images

Stage 1 – reflection in y = x sends (x, y) to (y, x). Its matrix is

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 5 & 5 \\ 1 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 3 & 5 & 5 \end{bmatrix}$$

Image A′B′C′: **A′(1, 3), B′(1, 5), C′(3, 5)**.

Stage 2 – enlargement, scale factor 2, centre O:

$$E = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}; \quad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 3 & 5 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 6 \\ 6 & 10 & 10 \end{bmatrix}$$

Image A″B″C″: **A″(2, 6), B″(2, 10), C″(6, 10)**.

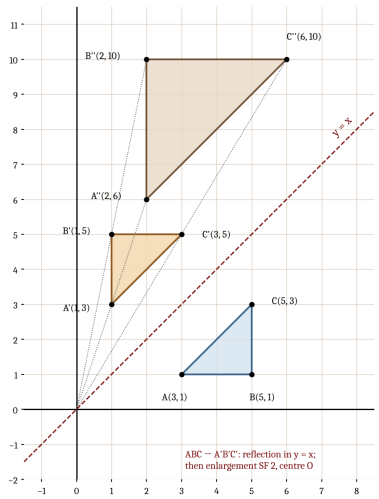


Figure 7(a): ABC, its reflection A′B′C′ and its enlargement A″B″C″

Single matrix for both stages (the first transformation is written on the right):

$$T = E\,M = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

Matrix that returns A″B″C″ to ABC is the inverse T^{-1}. The determinant of T is $(0 \times 0) - (2 \times 2) = -4$, so

$$T^{-1} = \frac{1}{-4} \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \tfrac{1}{2} \\ \tfrac{1}{2} & 0 \end{bmatrix}$$

Check: T^{-1} applied to A″(2, 6) gives $(\tfrac{1}{2} \times 6, \tfrac{1}{2} \times 2) = (3, 1) = A$ ✓.

CamScanner

∴ T = [[0, 2], [2, 0]] and the matrix that returns the image to ABC is T^{-1} = [[0, ½], [½, 0]].

(b) Areas of the transformed triangles

Area of ABC = ½ × base × height = ½ × 2 × 2 = 2 cm² (given). A reflection keeps the area, so area of A′B′C′ = **2 cm²**. Enlargement by scale factor 2 multiplies the area by $2^2 = 4$ (= |determinant of T|), so area of A″B″C″ = 4 × 2 = **8 cm²**.

∴ Areas: A′B′C′ = 2 cm² and A″B″C″ = 8 cm².

(c) Construction of triangle ABC and the position of D

Construction steps (using a ruler and a pair of compasses only):

1. Draw BC = 8 cm. 2. At B construct ∠ABC = 105° (90° + 15°, where 15° is got by bisecting 30°). 3. With centre C and radius 12 cm cut the arm at A, join AC.

4. Construct the perpendicular bisector of AB (every point on it is equidistant from A and B). 5. Bisect AC to get its midpoint M and draw the circle with diameter AC (the angle in a semicircle is 90°, so ∠ADC = 90° for any point D on it). 6. D is the intersection of the bisector and the circle, on the same side of AB as C.

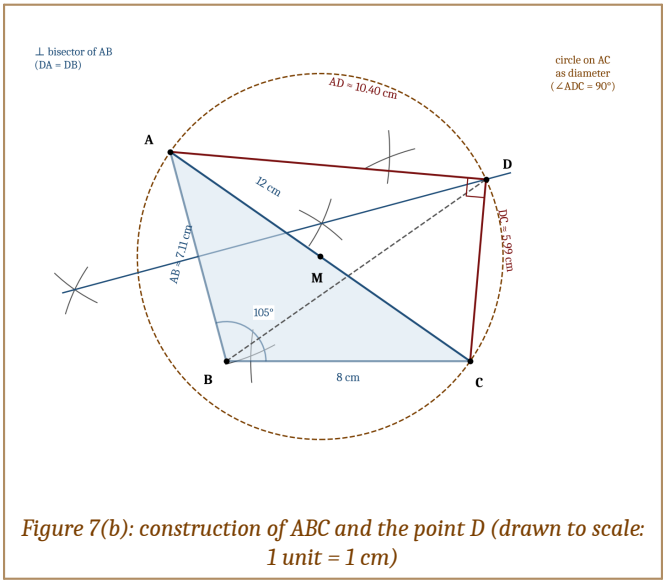


Figure 7(b): construction of ABC and the point D (drawn to scale: 1 unit = 1 cm)

Lengths (measured and checked by calculation):

$$\sin A = \frac{8 \sin 105^\circ}{12} = 0.6440 \implies \angle BAC = 40.09^\circ,\ \angle ACB = 180^\circ - 105^\circ - 40.09^\circ = 34.91^\circ$$

$$AB = \frac{12 \sin 34.91^\circ}{\sin 105^\circ} = 7.11 \text{ cm}$$

D lies on the perpendicular bisector of AB, so DA = DB. Measuring the construction gives ∠DAB = 70.0°, so ∠DAC = 70.0° − 40.09° = 29.9°. In the right-angled triangle ADC:

$$AD = 12 \cos 29.9^\circ = 10.40 \text{ cm},\ \ DC = 12 \sin 29.9^\circ = 5.99 \text{ cm}$$

∴ AB ≈ 7.1 cm, AD ≈ 10.4 cm and DC ≈ 6.0 cm (D is the point where the bisector of AB meets the circle on AC).

> **Key idea:** For matrices: the first transformation is written NEXT to the points. An enlargement of scale factor k multiplies areas by k^2; a reflection keeps lengths and areas.

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A SACCO makes saving boxes in the form of a pyramid lamp shed ABCDPQRS: the base ABCD is a square of side 10 cm, the top PQRS is a square of side 4 cm and each short edge is 5 cm. The box is filled with UGX 500 coins of diameter 2.2 cm and height 0.2 cm. The SACCO also wants a telecom mast on top of a hill whose base is 80 m from the SACCO office. The angle of elevation of the top of the hill from the office is 50° and of the top of the mast is 60°. The mast must not be more than 50 m.

Task: (a) Height of the saving box. (b) Money saved when the box is full. (c) Height of the hill. (d) Height of the mast, with a comment.

(a) Height of the saving box

The horizontal distance moved by a slant edge from the base corner to the top corner = half the difference of the diagonals:

$$\tfrac{1}{2}(10\sqrt{2} - 4\sqrt{2}) = 3\sqrt{2} = 4.243 \text{ cm}$$

In the right-angled triangle formed by the slant edge (5 cm), this horizontal run and the vertical height h:

$$h^2 = 5^2 - (3\sqrt{2})^2 = 25 - 18 = 7 \implies h = \sqrt{7} = 2.646 \text{ cm}$$

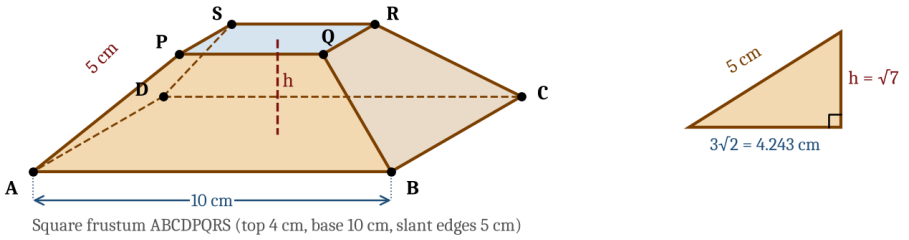


Figure 8(a): the frustum ABCDPQRS and the right-angled triangle that gives its height

∴ The height of the saving box is $\sqrt{7} \approx 2.65$ cm.

(b) Money saved when the box is full

Volume of the box = (big pyramid) − (small pyramid cut off). By similar pyramids, with H the height of the complete pyramid:

$$\frac{H - h}{H} = \frac{4}{10} \implies H = h \div 0.6 = 4.410 \text{ cm}, \text{ height of the small pyramid} = 4.410 - 2.646 = 1.764 \text{ cm}$$

$$V = \tfrac{1}{3} \times 10^2 \times 4.410 - \tfrac{1}{3} \times 4^2 \times 1.764 = 146.99 - 9.41 = 137.58 \text{ cm}^3$$

Volume of one coin $= \pi r^2 h = \pi \times 1.1^2 \times 0.2 = 0.7603 \text{ cm}^3$.

$$\text{Number of coins} = \frac{137.58}{0.7603} = 180.96 \implies 180 \text{ whole coins}$$

∴ The money saved when the box is full ≈ 180 × 500 = UGX 90,000 (an estimate, since coins leave small gaps).

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(c) Height of the hill and (d) Height of the mast

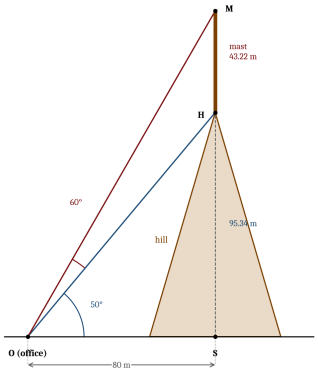


Figure 8(b): hill and mast seen from the SACCO office

$$\tan 50^\circ = \frac{\text{hill}}{80} \implies \text{hill} = 80 \times \tan 50^\circ = 80 \times 1.1918 = 95.34 \text{ m}$$

$$\tan 60^\circ = \frac{\text{hill} + \text{mast}}{80} \implies \text{hill} + \text{mast} = 80 \times 1.7321 = 138.56 \text{ m}$$

$$\text{mast} = 138.56 - 95.34 = 43.22 \text{ m}$$

$\therefore$ The hill is 95.34 m high and the mast is 43.22 m. Since 43.22 m < 50 m, the mast meets the safety (stability) limit and can be built.

> **Key idea:** Height of a frustum: use the right-angled triangle (slant edge, horizontal run between corners, height). For the mast subtract the two heights from the same observation point.

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QUESTION 9	SIMILAR SOLIDS, AREAS, CURRENCY EXCHANGE AND DEPRECIATION	SHURE

A water tank is a cylinder surmounted by a hemisphere of the same radius. The original tank has radius 3.5 m and volume 397.83 m^3. A similar larger tank has volume 1342.69 m^3 and its entire external surface is painted. Paint costs KSh 6,500 per drum and one drum paints 20 m^2. The officer goes to Kenya with UGX 32,000,000 (bank selling rate 1 KSh = UGX 135). A pickup truck bought at UGX 85,000,000 depreciates by 15% per annum; it will be sold when its value is UGX 52,200,625. (Use $\pi = 22/7$.)

Task: (a) Calculate the total area to be painted. (b) Find the number of drums and advise whether the money is enough. (c) How long should the truck be used before it is sold?

(a) Total area to be painted

Original tank. $V = \pi r^2 h + \frac{2}{3}\pi r^3$ with r = 3.5 m (the printed "cm^3" is read as m^3):

$$397.83 = 38.5h + 89.83 \implies h = 308 \div 38.5 = 8 \text{ m}$$

Scale factor.

$$k^3 = \frac{1342.69}{397.83} = 3.375 \implies k = \sqrt[3]{3.375} = 1.5$$

Larger tank: radius = 3.5 × 1.5 = 5.25 m, cylinder height = 8 × 1.5 = 12 m.

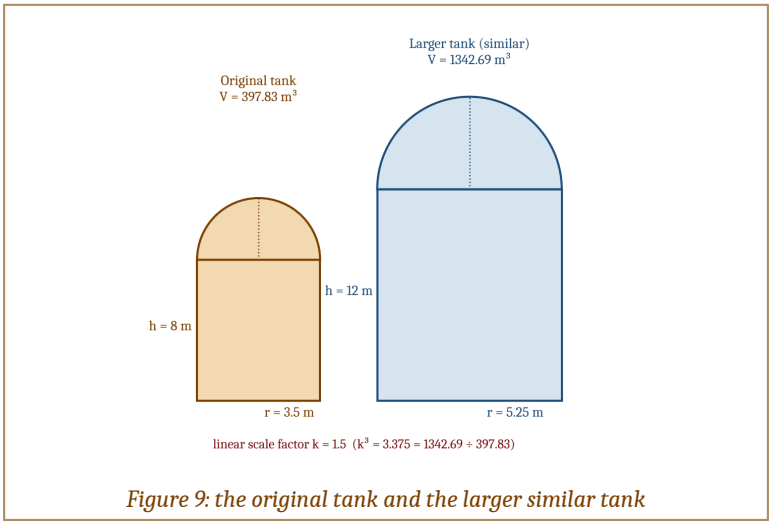


Figure 9: the original tank and the larger similar tank

External surface = curved wall of cylinder + base + curved surface of hemisphere:

$$2\pi rh = 2 \times 22/7 \times 5.25 \times 12 = 396 \text{ m}^2$$

$$\pi r^2 \text{ (base)} = 22/7 \times 5.25^2 = 86.625 \text{ m}^2$$

$$2\pi r^2 \text{ (dome)} = 2 \times 22/7 \times 5.25^2 = 173.25 \text{ m}^2$$

∴ Total area to be painted = 396 + 86.625 + 173.25 = 655.875 m^2.

(b) Drums of paint and the money available

$$\text{Drums} = \frac{655.875}{20} = 32.79 \implies 33 \text{ drums (round up)}$$

Cost in Kenya shillings = 33 × 6,500 = KSh 214,500. The officer is buying KSh from the bank, so the **bank selling rate** applies:

$$214{,}500 \times 135 = \text{UGX } 28{,}957{,}500$$

∴ The minimum is 33 drums. Since UGX 28,957,500 < UGX 32,000,000, the money is enough, with UGX 3,042,500 to spare.

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(c) Time to use the truck before selling

Value after n years = 85,000,000 × $(0.85)^n$ = 52,200,625, so $(0.85)^n$ = 52,200,625 ÷ 85,000,000 = 0.614125. By trial:

End of year	Working	Value (UGX)
1	85,000,000 × 0.85	72,250,000
2	72,250,000 × 0.85	61,412,500
3	61,412,500 × 0.85	52,200,625

∴ **The truck should be used for 3 years before it is sold.**

Key idea: Similar solids: $k^3 = V_2 \div V_1$. "Entire surface" includes the base. Buying foreign money from a bank uses the bank SELLING rate; round paint/tiles UP.

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Taking the district headquarters as the origin: a medical storage facility A is 2 km east and 4 km north; a health centre B is 14 km east and 10 km north. A main distribution centre P divides AB internally in the ratio 2 : 1. An alternate centre Q is the reflection of P in the line y = x. The chairman wants the area of triangle ABQ in hectares (10,000 m^2 = 1 ha).

Task: (a) Locations of P and Q from the headquarters. (b) The area of triangle ABQ in hectares.

(a) Locations of P and Q

A(2, 4), B(14, 10). Vector AB = (14 − 2, 10 − 4) = (12, 6). Since P divides AB in the ratio 2 : 1, AP = ⅔ of AB:

$$\text{AP} = \tfrac{2}{3}(12, 6) = (8, 4) \implies \text{OP} = \text{OA} + \text{AP} = (2 + 8, 4 + 4) = (10, 8)$$

Reflection in y = x interchanges the coordinates, so Q = (8, 10); in matrix form

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$$

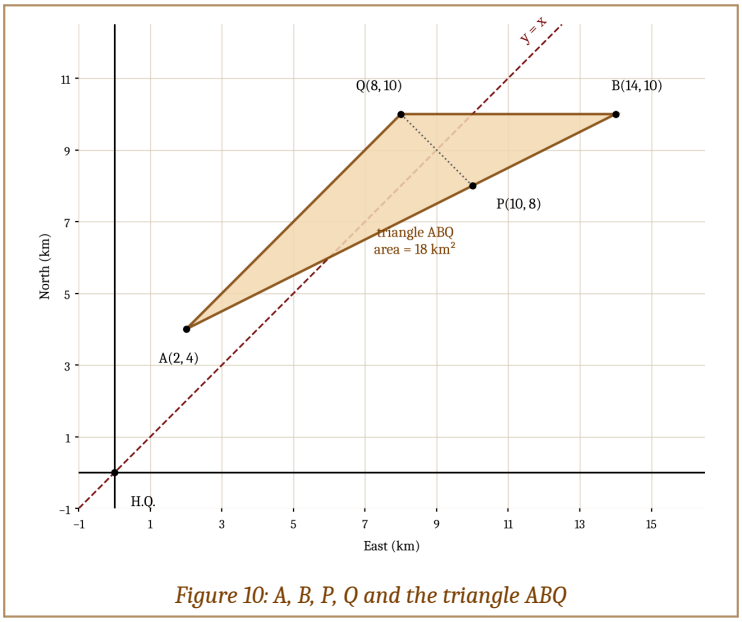


Figure 10: A, B, P, Q and the triangle ABQ

∴ P is 10 km east and 8 km north of the headquarters (12.8 km on a bearing of 051.3°); Q is 8 km east and 10 km north (12.8 km on 038.7°).

(b) Area of triangle ABQ in hectares

B(14, 10) and Q(8, 10) have the same y-coordinate, so BQ is horizontal: BQ = 14 − 8 = 6 km. The perpendicular height from A(2, 4) to the line y = 10 is 10 − 4 = 6 km.

$$\text{Area} = \tfrac{1}{2} \times \text{base} \times \text{height} = \tfrac{1}{2} \times 6 \times 6 = 18 \text{ km}^2$$

$$18 \text{ km}^2 = 18 \times 1{,}000{,}000 \text{ m}^2 = 18{,}000{,}000 \text{ m}^2 \implies 18{,}000{,}000 \div 10{,}000 = 1{,}800 \text{ ha}$$

∴ The triangular region has an area of 1,800 hectares.

Key idea: Internal division 2 : 1 means P is two-thirds of the way from A to B. Reflection in y = x swaps x and y. Use a horizontal side as the base for an easy area.

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A container's interior measures 6 ft by 20 ft by 5 ft. A generator has a rectangular base with corners at (−4, 3), (−1, 6), (−3, 8) and (−6, 5). A crane fixed at one place, using only its rotating arm, shifts the corners to (3, 6), (6, 3), (8, 5) and (5, 8) respectively.

Task: (a) Find (i) the length of the longest pipe that fits in the container and (ii) the angle it makes with the floor. (b) Find (i) the position where the crane is fixed and (ii) the angle through which the arm rotates.

(a) Longest pipe and its angle with the floor

The longest pipe is the **space diagonal** AD of the cuboid. Floor: AB = 20 ft, BC = 6 ft; height CD = 5 ft.

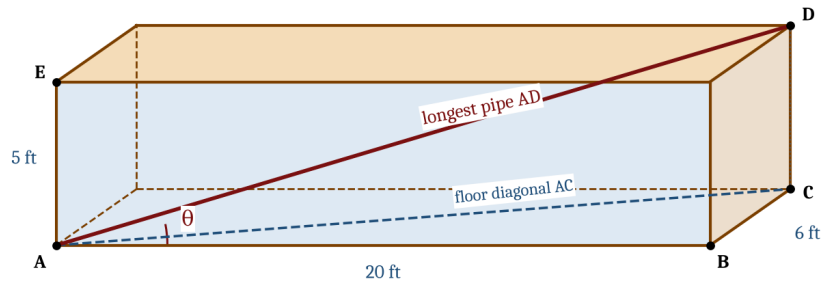


Figure 11(a): the container with the floor diagonal AC and the longest pipe AD

$$AC^2 = 20^2 + 6^2 = 436 \implies AC = 20.88 \text{ ft}$$

$$AD^2 = AC^2 + CD^2 = 436 + 25 = 461 \implies AD = \sqrt{461} = 21.47 \text{ ft}$$

The angle between the pipe AD and the floor is $\angle DAC = \theta$ in the right-angled triangle ACD:

$$\tan\theta = \frac{CD}{AC} = \frac{5}{20.88} = 0.2395 \implies \theta = 13.47°$$

∴ The longest pipe is 21.47 ft and it makes an angle of 13.5° with the floor.

(b) The crane: centre and angle of rotation

The centre of rotation lies on the **perpendicular bisector** of the line joining any point to its image. Take A(−4, 3) → A′(3, 6) and B(−1, 6) → B′(6, 3).

Bisector of AA′: midpoint (−0.5, 4.5); gradient of AA′ = 3/7, so the perpendicular gradient = −7/3.

$$y - 4.5 = -\frac{7}{3}(x + 0.5) \implies 7x + 3y = 10 \quad \ldots (1)$$

Bisector of BB′: midpoint (2.5, 4.5); gradient of BB′ = −3/7, so the perpendicular gradient = 7/3.

$$y - 4.5 = \frac{7}{3}(x - 2.5) \implies 7x - 3y = 4 \quad \ldots (2)$$

Adding (1) and (2): 14x = 14, so x = 1. From (1): 7 + 3y = 10, so y = 1. The centre is **X(1, 1)**.

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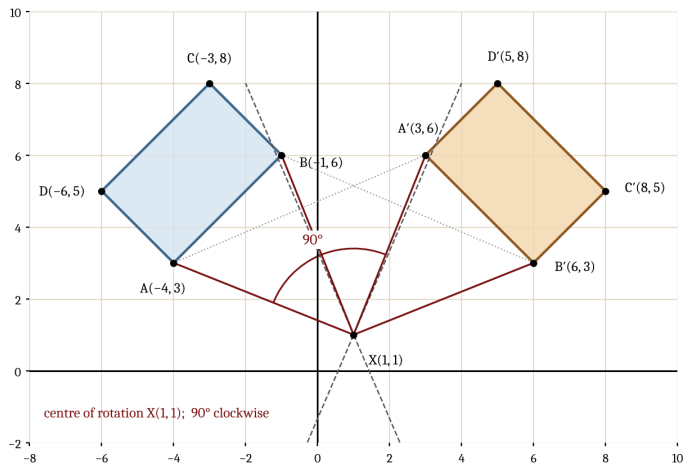


Figure 11(b): the generator base, its image and the centre of rotation X(1, 1)

Angle: XA = (−4 − 1, 3 − 1) = (−5, 2) and XA′ = (3 − 1, 6 − 1) = (2, 5). Their dot product is (−5)(2) + (2)(5) = 0, so the angle between them is 90°. XA points to the upper left and XA′ to the upper right, so the turn is **clockwise**. (Check: the rule (x, y) → (y, 2 − x) sends A(−4, 3) to (3, 6) ✓ and B(−1, 6) to (6, 3) ✓.)

∴ **The crane should be fixed at (1, 1) and its arm rotates through 90° clockwise.**

Key idea: Longest object in a box = space diagonal: $\sqrt{(l^2 + w^2 + h^2)}$. The centre of a rotation is where the perpendicular bisectors of point–image pairs meet.

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QUESTION 12 CIRCLE PROPERTIES AND MORTGAGE UTES

A neighbour built a circular tank base of radius 10 ft whose centre is 8 ft from the straight boundary of Mr Kintu's land, so part of it entered his land. Mr Kintu also buys a car at Shs 20,000,000 using a 5-year mortgage loan from Centenary Bank at 12% per annum, to be cleared in equal monthly instalments.

Task: Help Mr Kintu to determine (a) the area of his land that the neighbour's tank base entered; (b) the total interest for his loan.

(a) Area of the land entered by the tank base

The part inside Mr Kintu's land is a **segment** cut off by the boundary (chord PQ), at distance 8 ft from the centre O. In the right-angled triangle formed by the radius, the perpendicular 8 ft and half the chord:

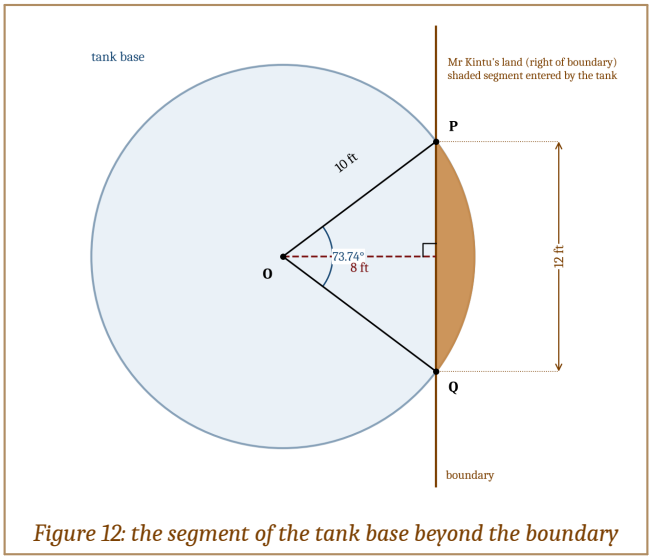


Figure 12: the segment of the tank base beyond the boundary

$$\cos\frac{\theta}{2} = \frac{8}{10} = 0.8 \Longrightarrow \theta/2 = 36.87^\circ \Longrightarrow \theta = 73.74^\circ$$

$$\text{half chord} = \sqrt{(10^2 - 8^2)} = 6 \text{ ft} \Longrightarrow \text{chord PQ} = 12 \text{ ft}$$

$$\text{Sector} = \frac{73.74}{360} \times \pi \times 10^2 = 64.35 \text{ ft}^2, \quad \text{Triangle OPQ} = \tfrac{1}{2} \times 12 \times 8 = 48 \text{ ft}^2$$

$$\text{Segment} = \text{sector} - \text{triangle} = 64.35 - 48 = 16.35 \text{ ft}^2$$

∴ The area of Mr Kintu's land entered by the tank base is ≈ 16.35 ft².

(b) Total interest on the mortgage

Principal P = Shs 20,000,000; monthly rate i = 12% ÷ 12 = 1% = 0.01; number of months n = 5 × 12 = 60. The mortgage (equal monthly instalment) formula is

$$M = \frac{P \times i}{1 - (1+i)^{-n}} = \frac{20{,}000{,}000 \times 0.01}{1 - (1.01)^{-60}} = \frac{200{,}000}{1 - 0.55045} = \frac{200{,}000}{0.44955} = \text{Shs } 444{,}889$$

Total repaid = 444,889 × 60 = Shs 26,693,337 (using the unrounded instalment).

Total interest = 26,693,337 − 20,000,000 = Shs 6,693,337.

∴ The monthly instalment is about Shs 444,889 and the total interest on the loan is ≈ Shs 6.69 million (Shs 6,693,337).

Key idea: A segment = sector − triangle. The 6-8-10 triangle appears here, so no calculator is needed for the half chord. Mortgage: use the monthly rate and the number of MONTHS.

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QUESTION 13	VECTOR GEOMETRY, BEARING, SURFACE AREA AND VOLUME	UMTA

A council maps three control points relative to the post office P: A is 6 km East and 8 km North of P; B is 14 km east and 2 km north; C is 5 km west and 7 km north. The central pavilion is a hollow right circular cone on top of a solid cylinder of diameter 30 m and height 12 m; the base angle of the cone is 35°. The exterior walls are weatherproofed and the inside is fitted with a climate-control system.

Task: (a) State the direct distances and bearings of B and C from A. (b) Calculate (i) the area to be weatherproofed and (ii) the volume of air space inside the pavilion.

(a) Distances and bearings of B and C from A

Taking P as the origin: A(6, 8), B(14, 2), C(−5, 7).

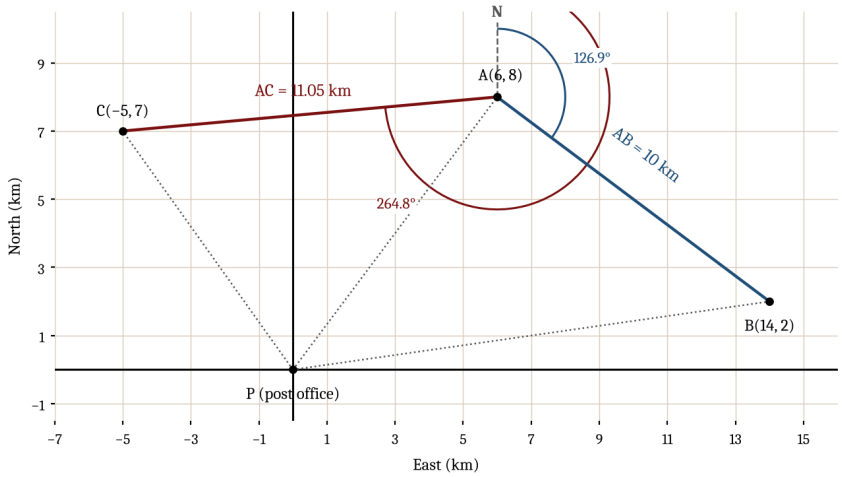


Figure 13(a): the control points and the bearings from A

B from A: AB = B − A = (14 − 6, 2 − 8) = (8, −6).

$$|AB| = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10 \text{ km}$$

B is 8 km east and 6 km south of A, so the angle measured from south = $\tan^{-1}(8 \div 6) = 53.13°$ (S53.13°E). Bearing = 180° − 53.13° = **126.87°**.

C from A: AC = C − A = (−5 − 6, 7 − 8) = (−11, −1).

$$|AC| = \sqrt{((-11)^2 + (-1)^2)} = \sqrt{122} = 11.05 \text{ km}$$

C is 11 km west and 1 km south of A: angle from south = $\tan^{-1}(11 \div 1) = 84.80°$ (S84.8°W). Bearing = 180° + 84.80° = **264.8°**.

∴ Directory: B is 10 km from A on a bearing of 126.9°; C is 11.05 km from A on a bearing of 264.8°.

(b) Weatherproofing and air space

Radius r = 30 ÷ 2 = 15 m; cylinder height H = 12 m. In the cone, the angle between the slant side and the base is 35°, so

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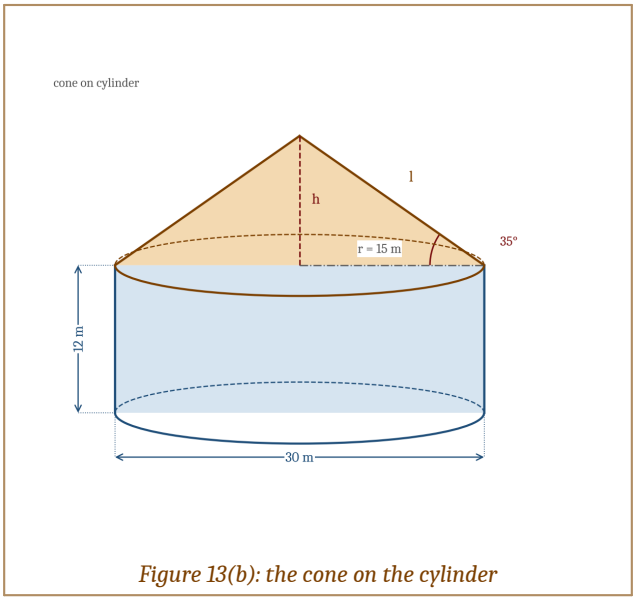


Figure 13(b): the cone on the cylinder

cone height h = 15 tan 35° = 10.50 m, slant length l = 15 ÷ cos 35° = 18.31 m

(i) Exterior walls = curved surface of the cylinder + curved surface of the cone:

$$2\pi rH = 2 \times \pi \times 15 \times 12 = 1{,}130.97 \text{ m}^2$$

$$\pi rl = \pi \times 15 \times 18.31 = 862.9 \text{ m}^2$$

∴ Area to be weatherproofed = 1,130.97 + 862.9 = ≈ 1,994 m².

(ii) Volume of air space = cylinder + cone:

$$\pi r^2 H = \pi \times 15^2 \times 12 = 8{,}482.3 \text{ m}^3 \;;\; \tfrac{1}{3}\pi r^2 h = \tfrac{1}{3} \times \pi \times 15^2 \times 10.50 = 2{,}474.7 \text{ m}^3$$

∴ Volume of the pavilion = 8,482.3 + 2,474.7 = ≈ 10,957 m³.

Key idea: Bearing of a point = measure clockwise from North. Build a right-angled triangle first (east/west and north/south) then add 90°, 180° or 270° as needed.

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A camping zone ABCD: from A the boundary runs 60 m on a bearing of 045° to B, then 80 m on a bearing of 135° to C. A security outpost is placed equidistant from the boundary lines AB and BC and equidistant from A and C. A triangular field ACD has CD = 110 m and ∠ACD = 80°. An underground cable runs from A to D at Shs 166,500 per metre. The quantity surveyor earns a gross monthly pay of Shs 11,523,000 taxed with PAYE: 0 – 335,000: 0%; 335,001 – 410,000: 10%; 410,001 – 485,000: 20%; 485,001 – 1,000,000: 30%; above 1,000,000: 40%.

Task: (a) Using construction, (i) find the actual distance and bearing of the outpost from A; (ii) calculate the exact cost of cable AD. (b) Calculate the surveyor's net monthly pay.

(a)(i) The security outpost

At B the back-bearing of BA is 045° + 180° = 225° and BC is on 135°, so ∠ABC = 225° − 135° = **90°**. Also AC = $\sqrt{(60^2 + 80^2)}$ = **100 m**.

Construction (scale 1 cm : 10 m): draw AB = 6 cm on 045° from A; at B draw BC = 8 cm on 135°; join AC. **(1)** Bisect ∠ABC (points equidistant from AB and BC). **(2)** Construct the perpendicular bisector of AC (points equidistant from A and C). **(3)** Their intersection is the outpost O.

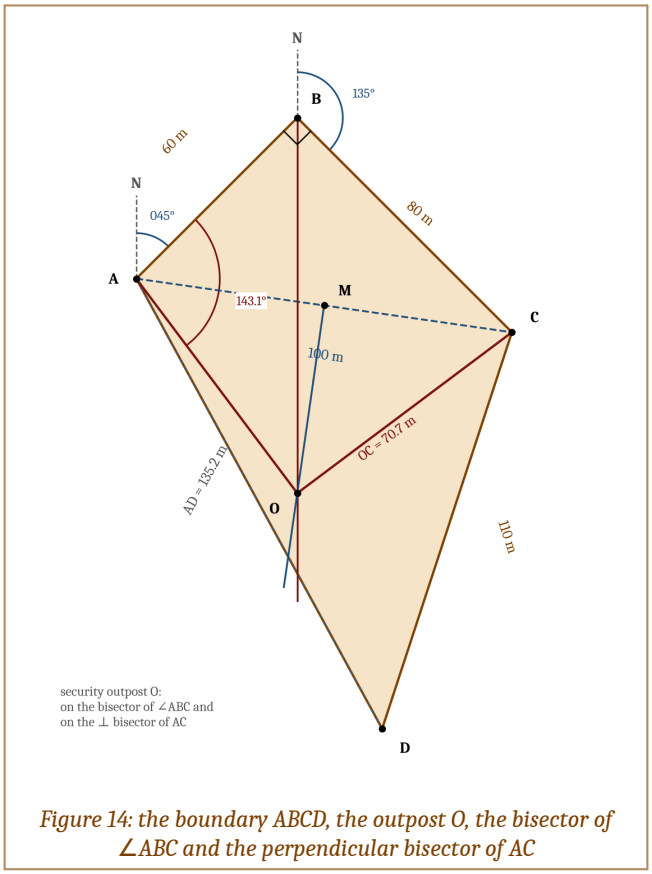


Figure 14: the boundary ABCD, the outpost O, the bisector of ∠ABC and the perpendicular bisector of AC

From the scale drawing, AO ≈ 7.07 cm, i.e. 70.7 m, on a bearing of about 143°. **Check by calculation:** the bisector of the right angle at B meets the perpendicular bisector of the hypotenuse at the midpoint of arc AC, so triangle AOC is right-angled and isosceles:

$$AO = OC = \frac{AC}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.7 \text{ m}, \quad \angle OAC = 45°$$

Bearing of C from A = 90° + $\tan^{-1}(14.14 \div 99)$ = 98.13°; O is on the opposite side of AC from B, so the bearing of O = 98.13° + 45° = 143.13°.

∴ The outpost is 70.7 m from A on a bearing of 143°.

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(a)(ii) Cost of the cable AD

Triangle ACD: AC = 100 m, CD = 110 m, ∠ACD = 80°. Cosine rule:

$$AD^2 = 100^2 + 110^2 - 2(100)(110)\cos 80° = 22{,}100 - 22{,}000 \times 0.17365 = 22{,}100 - 3{,}820.3 = 18{,}279.7$$

$$AD = \sqrt{18{,}279.7} = 135.2 \text{ m}$$

$$\text{Cost} = 135.2 \times 166{,}500 = \text{Shs } 22{,}510{,}800$$

∴ The cable AD costs ≈ Shs 22,510,800.

(b) Surveyor's net monthly pay

Band of income (Shs)	Amount in the band	Rate	Tax (Shs)
0 – 335,000	335,000	0%	0
335,001 – 410,000	75,000	10%	7,500
410,001 – 485,000	75,000	20%	15,000
485,001 – 1,000,000	515,000	30%	154,500
Above 1,000,000	11,523,000 – 1,000,000 = 10,523,000	40%	4,209,200
Total PAYE			**4,386,200**

$$\text{Net pay} = \text{gross pay} - \text{PAYE} = 11{,}523{,}000 - 4{,}386{,}200 = \text{Shs } 7{,}136{,}800$$

∴ The surveyor's net monthly pay is Shs 7,136,800.

> **Key idea:** Tax is charged band by band (only the part of income inside a band is taxed at that rate). A right angle at B makes the outpost the midpoint of the arc AC, giving a quick check of the scale drawing.

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Jessica plans to use a room as a beauty salon. She has 4.7 m^2 of wallpaper and the wall to cover is 240 cm by 300 cm. A roll of 10 m^2 of wallpaper is sold at Shs 32,000. Her salon logo is triangle ABC with A(1, 1), B(4, 1), C(1, 5). She reflects it in the x-axis to get A′B′C′, then enlarges A′B′C′ with scale factor 2 about the origin to get A″B″C″.

Task: (a) Determine the extra roll she must buy. (b) Determine the exact coordinates of the two images. (c) Describe a single geometric transformation that maps ABC directly onto A″B″C″.

(a) Extra wallpaper to buy

Area of the wall = 240 cm × 300 cm = 72,000 cm^2 = 7.2 m^2 (2.4 m × 3.0 m).

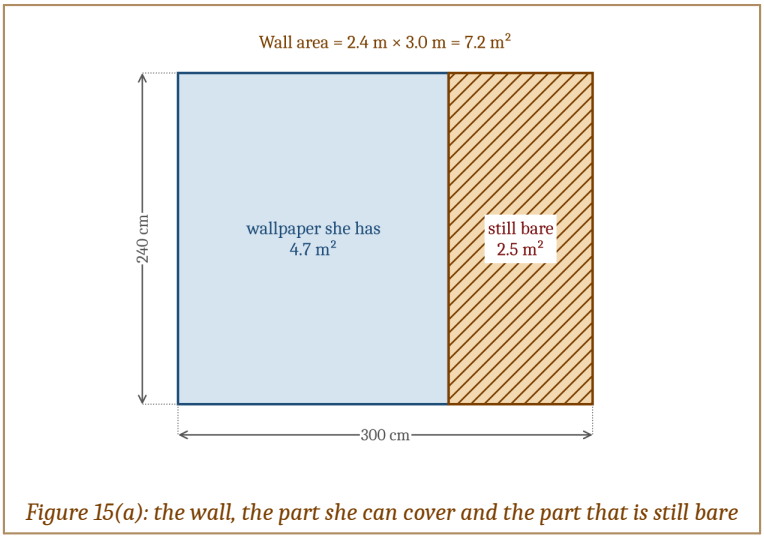


Figure 15(a): the wall, the part she can cover and the part that is still bare

Extra needed = 7.2 − 4.7 = 2.5 m^2

Wallpaper is sold only in rolls of 10 m^2, so she must buy **1 roll**, which costs **Shs 32,000** (about 7.5 m^2 remains unused).

∴ **Jessica must buy one more roll (10 m^2) at Shs 32,000.**

(b) Coordinates of the images

Reflection in the x-axis sends (x, y) to (x, −y). Matrix R:

$$R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix};\ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 4 & 1 \\ 1 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 1 \\ -1 & -1 & -5 \end{bmatrix}$$

A′(1, −1), B′(4, −1), C′(1, −5).

Enlargement, scale factor 2, centre O (matrix E):

$$E = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix};\ \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 4 & 1 \\ -1 & -1 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 8 & 2 \\ -2 & -2 & -10 \end{bmatrix}$$

A″(2, −2), B″(8, −2), C″(2, −10).

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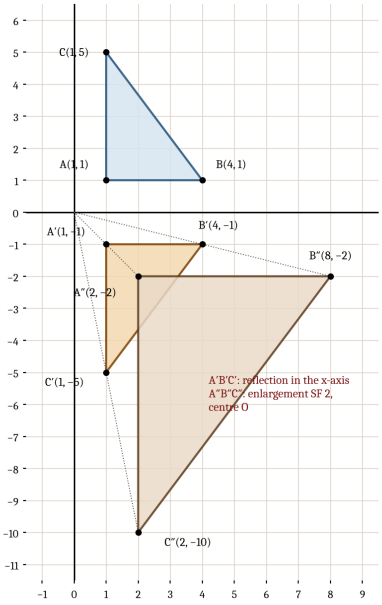


Figure 15(b): ABC, A′B′C′ and A″B″C″

∴ A′(1, –1), B′(4, –1), C′(1, –5) and A″(2, –2), B″(8, –2), C″(2, –10).

(c) A single transformation from ABC to A″B″C″

Combine the two matrices (reflection first, then enlargement):

$$T = E\,R = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

This single matrix sends (x, y) to **(2x, –2y)**. Check: A(1, 1) → (2, –2) = A″ ✓; C(1, 5) → (2, –10) = C″ ✓. Because the determinant is (2)(–2) – 0 = –4, the shape is **enlarged by scale factor 2 (areas × 4) and reversed in orientation**. Area of ABC = ½ × 3 × 4 = 6 square units, so area of A″B″C″ = 24 square units.

∴ The single transformation is described by the matrix [[2, 0], [0, –2]]: an enlargement of scale factor 2, centre the origin, combined with a reflection in the x-axis (no single pure reflection or pure enlargement can do this).

> **Key idea:** Multiply the matrices in the ORDER of the transformations (second one on the left). Wallpaper, tiles and paint are bought in whole units.

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ANSWERS AT A GLANCE – QUESTIONS 1 TO 15

Q	Final answers
1	(a) common area ≈ 55.2 m^2 (b) 29.15 m (return on ≈ 299°)
2	(a) Bank B (interest Shs 503,328 against Shs 672,000) (b) option (i): Shs 35,000,000 against Shs 36,760,000
3	common area 0.626 m^2 (6,258 cm^2) → 191 pavers → Shs 47,750
4	(a) height 8 cm, diameter 11.06 cm (b) 42,210 cm^2 of material; cost Shs 50,000 (c) Shs 1,600 per cone; profit Shs 350,000
5	(a) 163.1 km (b) back at about 4:13 p.m. (before 5:30 p.m.) – no overtime
6	(a) 39.55 cm (b) 1,898 cm^3 > 1,500 cm^3 – enough space (c) Shs 2,968,617 per instalment
7	A′(1,3) B′(1,5) C′(3,5); A″(2,6) B″(2,10) C″(6,10); T = [[0,2],[2,0]]; T^{-1} = [[0,½],[½,0]]; areas 2 cm^2 and 8 cm^2; AB ≈ 7.1 cm, AD ≈ 10.4 cm, DC ≈ 6.0 cm
8	(a) $\sqrt{7}$ ≈ 2.65 cm (b) 180 coins = UGX 90,000 (c) hill 95.34 m (d) mast 43.22 m (< 50 m, acceptable)
9	(a) 655.875 m^2 (b) 33 drums, KSh 214,500 = UGX 28,957,500 – enough (c) 3 years
10	(a) P(10, 8), Q(8, 10) (b) 18 km^2 = 1,800 hectares
11	(a) 21.47 ft; 13.47° (b) centre (1, 1); 90° clockwise
12	(a) 16.35 ft^2 (b) interest ≈ Shs 6,693,337 (monthly instalment ≈ Shs 444,889)
13	(a) B: 10 km on 126.9°; C: 11.05 km on 264.8° (b) ≈ 1,994 m^2; ≈ 10,957 m^3
14	(a) outpost 70.7 m on 143°; cable cost ≈ Shs 22,510,800 (b) net pay Shs 7,136,800
15	(a) 1 roll (Shs 32,000) (b) A′(1,–1) B′(4,–1) C′(1,–5); A″(2,–2) B″(8,–2) C″(2,–10) (c) matrix [[2,0],[0,–2]]

Wishing all the candidates success in the coming UNEB examinations.

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